CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

Learner name / date	Lacey Merced 11/2/2026
Import status	20 rows loaded

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03, R14
Q02	Work order is unique	2	10%	R05, R06
Q03	Status uses the agreed list	2	10%	R08, R16
Q04	Duration is in range	3	15%	R10, R18, R20
Q05	READY identifies an inspector	3	15%	R03. R07, R12
Distinct rows with any issue		11	55%	Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

r03 FAILS q01 BECAUSE Machine_ID is blank. W07-S02 says Machine_ID is required. Without it the ispection record can't clearly identify which machine the info belongs to.

Next action: what to verify, who owns it, and how to retest

The Maintenance Records Owner should check the original inspection record and verify the correctr Machine_ID. After it's confirmed Q01 should e tested again.

Limit: why you cannot guess a correction or prove accuracy

I can't guess the missing Machine_ID because the spreadsheet doesn't prove what the correct value should be. A record that passes the spreadsheet rules also doesn't prove what actually happened to the machine.

Professional connection: one framework idea and how you used it

I used the fit-for-purpose idea from the Data Quality Framework. Data should contain information needed for its intended use. In this assignment a machine inspection record needs a Machine_ID so the ispection can be connected to the correct machine.

Refresh test: before, changed value, after, and restored result

Before Q04 count:2. Changed R01 Inspection_Minutes from 20 to) and refreshed. That made Q04 increase to 3. Restored R01 back to 20 and refresed again and Q04 changed back to 2.

Export only Report: two landscape pages, page break before row 26. Review both pages.